

LOW DROPOUT VOLTAGE REGULATOR

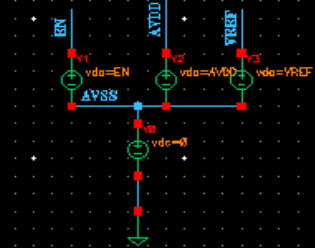
CADENCE LAB REPORT

TANUSHREE L

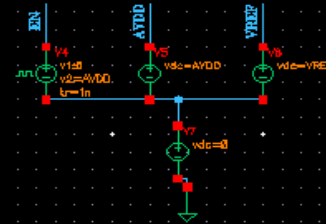
LDO CIRCUIT DESIGN SPECIFICATIONS

- **OUTPUT VOLTAGE : 1.2V**
- **LOAD CURRENT: 8m A**
- **REFERENCE VOLTAGE:1V**
- **IQ CURRENT: 60uA**
- **TECHNOLOGY: UMC 28nm**

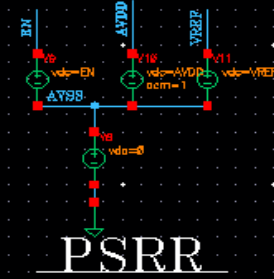
TEST BENCH



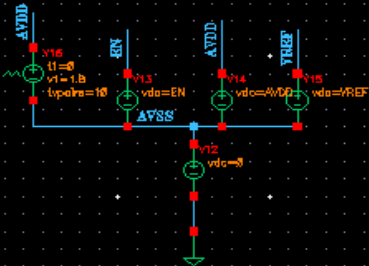
GAIN, PM, BANDWIDTH



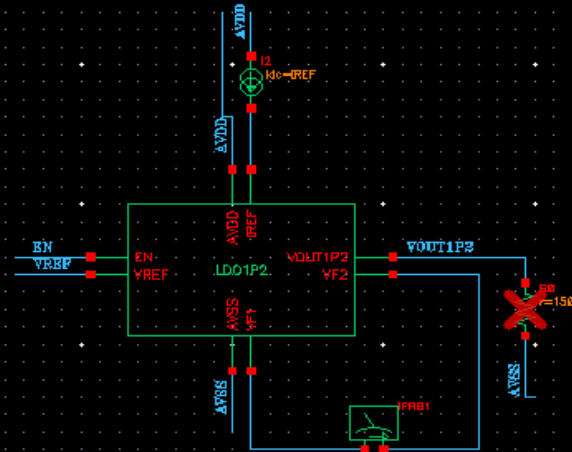
TRANSIENT



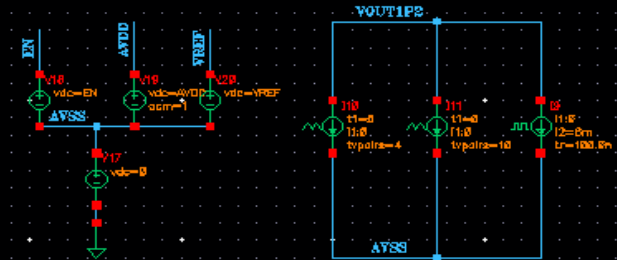
PSRR



LINE REGULATION

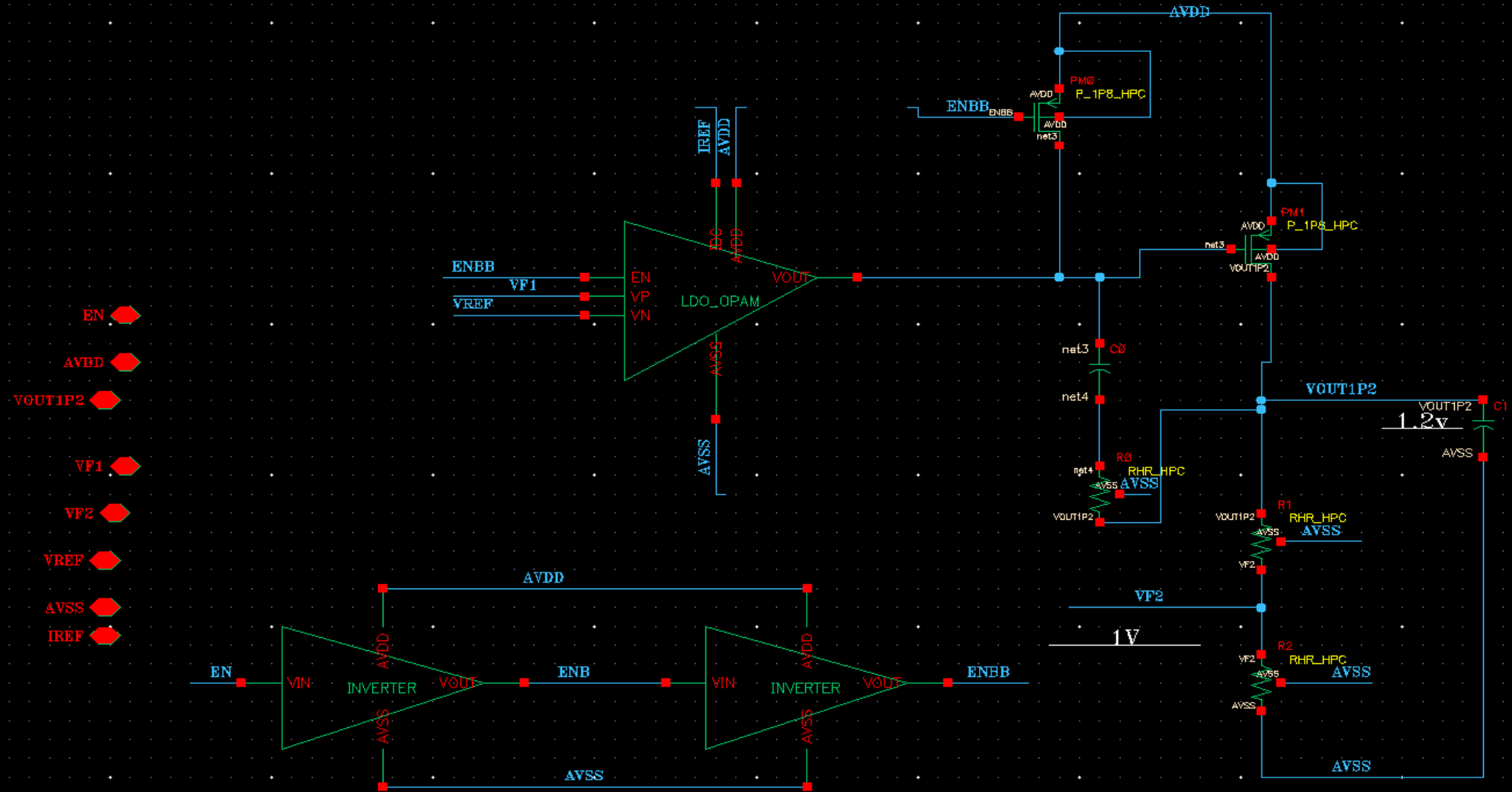


LDO



LOAD REGULATION

LDO CORE

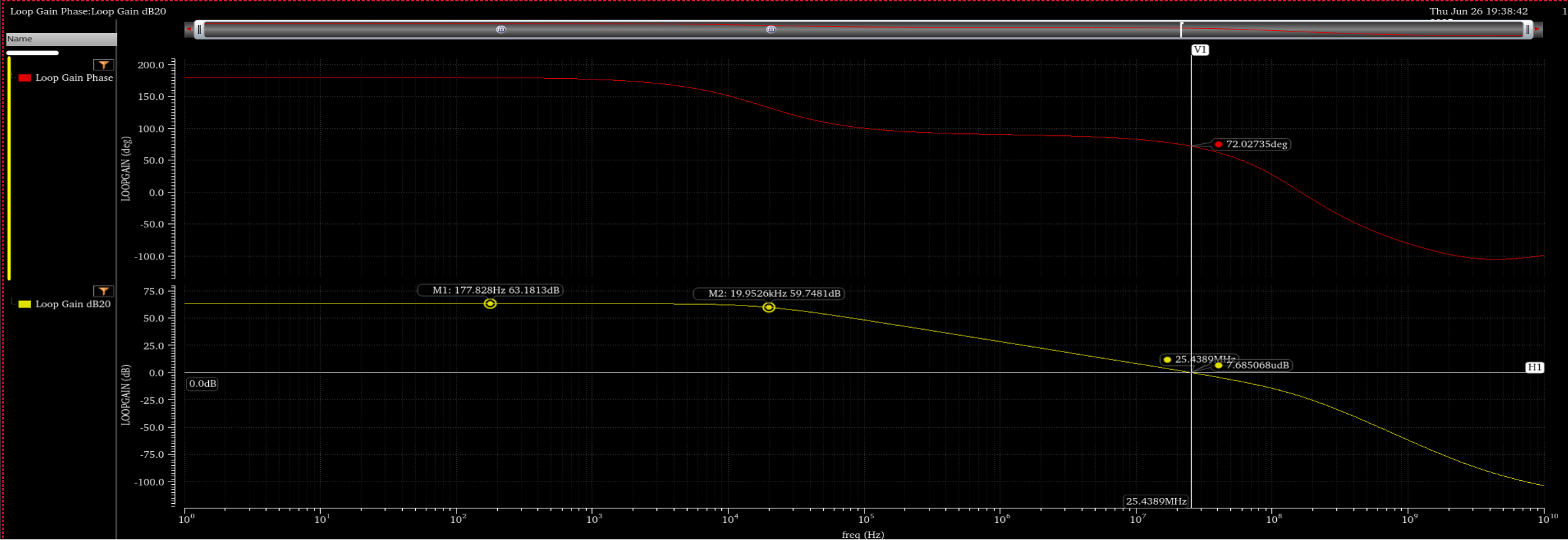


S.NO		MIN	TYPICAL	MAX
1	Output Voltage No load(V)	1.2000	1.2007	1.2020
2	Output Voltage Full load(V)	1.1956	1.1967	1.1969
3	Bandwidth No load(Hz)	151.3K	161.3K	212.6K
4	Bandwidth Full load(Hz)	268.4K	334K	479.1K
5	Gain No Load(dB)	56.24	63.18	64.18
6	Gain Full Load (dB)	46.71	50.35	50.79
7	Phase margin No load(deg)	70.57	72.04	73.83
8	Phase margin Full load(deg)	88.06	89.85	91.16
9	No load to full load error(%)	0.36	0.33	0.42
10	DC PSRR No load(dB)	-91.9734	-97.9734	-72.0234
11	PSRR at 10MHz No Load	-10.3345	-9.0726	-7.2335
12	DC PSRR Full load(dB)	-52.4504	-50.1063	-44.5266
13	PSRR at 10MHz Full load	-9.73013	-9.17399	-7.89895
14	Load regulation(%)	0.0106	0.0120	0.0167
15	Line regulation No load	0.003673	0.003673	0.02039
16	Line regulation Full load	3.2147	3.3727	3.3727

GAIN ,PHASE MARGIN AND BANDWIDTH –NO LOAD

SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
Gain(dB) min	56.24	125	FF	1.98
Gain(dB) max	64.18	-40	SS	1.62
BW(Hz) min	151.3K	25	TT	1.8
BW(Hz) max	212.6K	125	FF	1.98
PM(deg) min	70.58	125	FF	1.98
PM(deg)max	73.83	-40	SS	1.62
Vout(V) min	1.200	-40	SS	1.62
Vout (V) max	1.202	125	FF	1.98
Leakage current(A) min	56.46u	-40	SS	1.62
Leakage current(A) max	63.85u	125	FF	1.98

GAIN,PM AND BANDWIDTH :NO LOAD (TYPICAL)

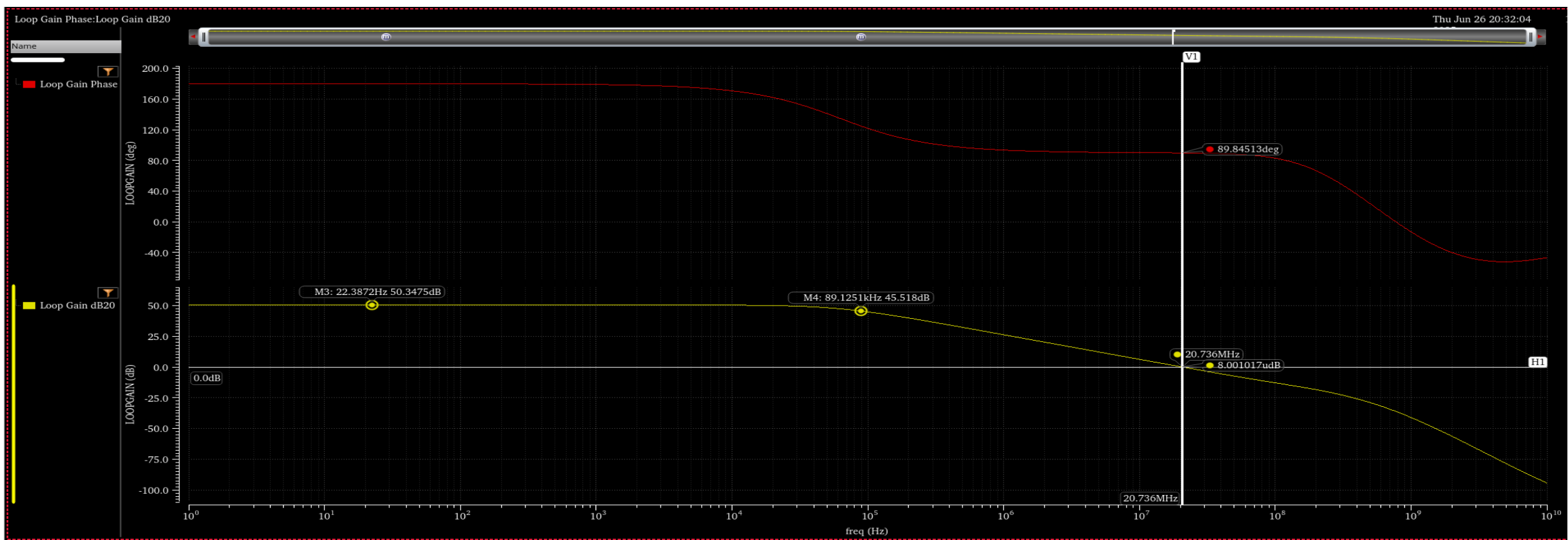


Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	SS	FF
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Ido_LDO_TB_1	Phase Margin				70.58	73.83	72.03	73.83	70.58
Ido_LDO_TB_1	bandwidth(db(...				151.3K	212.6K	151.3K	161.3K	212.6K
Ido_LDO_TB_1	value(db(mag(g...)				56.25	64.18	63.18	64.18	56.25
Ido_LDO_TB_1	OP("/I1/PM1" "r...				3	3	3	3	3
Ido_LDO_TB_1	VDC("/VOUT1P2")				1.2	1.202	1.201	1.2	1.202
Ido_LDO_TB_1	IDC("/I1/PM1/S")				56.46u	63.85u	60.03u	56.46u	63.85u

GAIN ,PHASE MARGIN AND BANDWIDTH –FULL LOAD

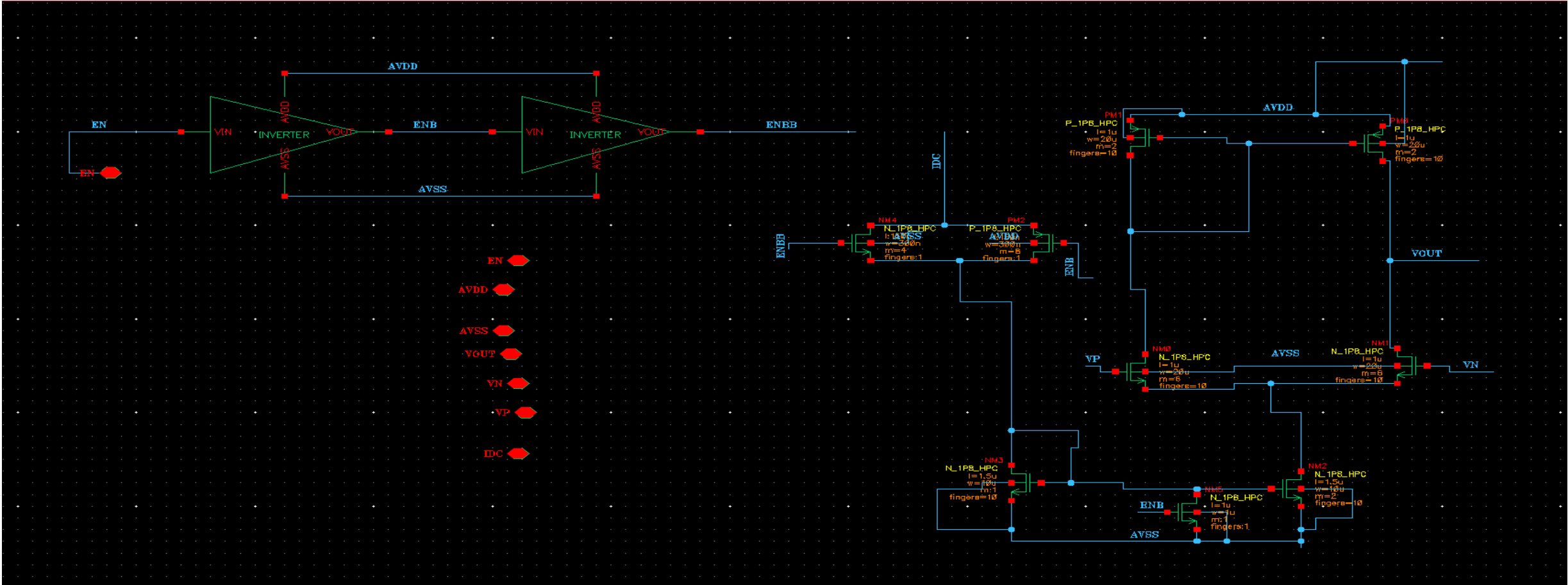
SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
Gain(dB) min	46.71	-40	SS	1.62
Gain(dB) max	50.79	125	FF	1.98
BW(Hz) min	268.3K	125	FF	1.98
BW(Hz) max	479.1K	-40	SS	1.62
PM(deg) min	88.06	125	FF	1.98
PM(deg)max	91.16	-40	SS	1.62
Vout(V) min	1.196	-40	SS	1.62
Vout (V) max	1.197	125	FF	1.98
Leakage current(A) min	8.027m	-40	SS	1.62
Leakage current(A) max	8.043m	125	FF	1.98

GAIN,PM AND BANDWIDTH :FULL LOAD (TYPICAL)



Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	SS	FF
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
Ido_LDO_TB_1	Phase Margin				88.06	91.16	89.85	91.16	88.06
Ido_LDO_TB_1	bandwidth(db(...				268.3K	479.1K	333.8K	479.1K	268.3K
Ido_LDO_TB_1	value(db(mag(g...				46.71	50.79	50.35	46.71	50.79
Ido_LDO_TB_1	OP("/I1/PM1" "r...				2	2	2	2	2
Ido_LDO_TB_1	VDC("/VOUT1P2")				1.196	1.197	1.197	1.196	1.197
Ido_LDO_TB_1	IDC("/I1/PM1/S")				8.027m	8.043m	8.037m	8.027m	8.043m

OPAMP SIMULATION



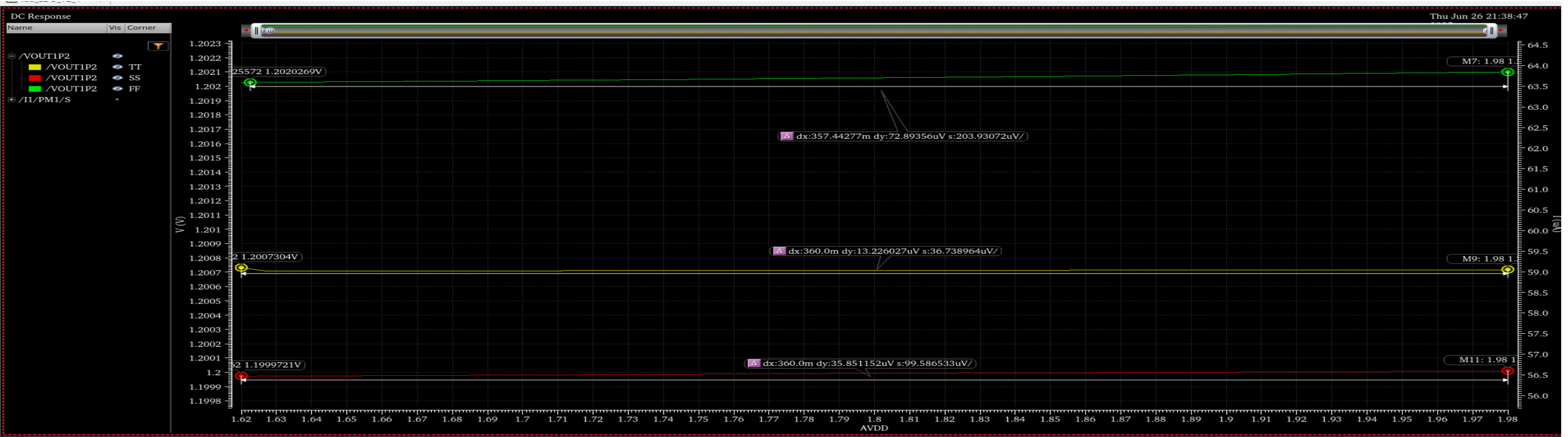
Test	Output	Spec	Weight	Pass/Fail	Min	Max	TT	SS	FF
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
single_stage_op...	/VP								
single_stage_op...	/VOUT								
single_stage_op...	Phase Margin				87.79	87.84	87.79	87.84	87.83
single_stage_op...	bandwidth(db(...				153.7K	164K	158.4K	153.7K	164K
single_stage_op...	value(db(mag(g...				38.22	41.24	40.18	41.24	38.22

LINE REGULATION : FULL LOAD



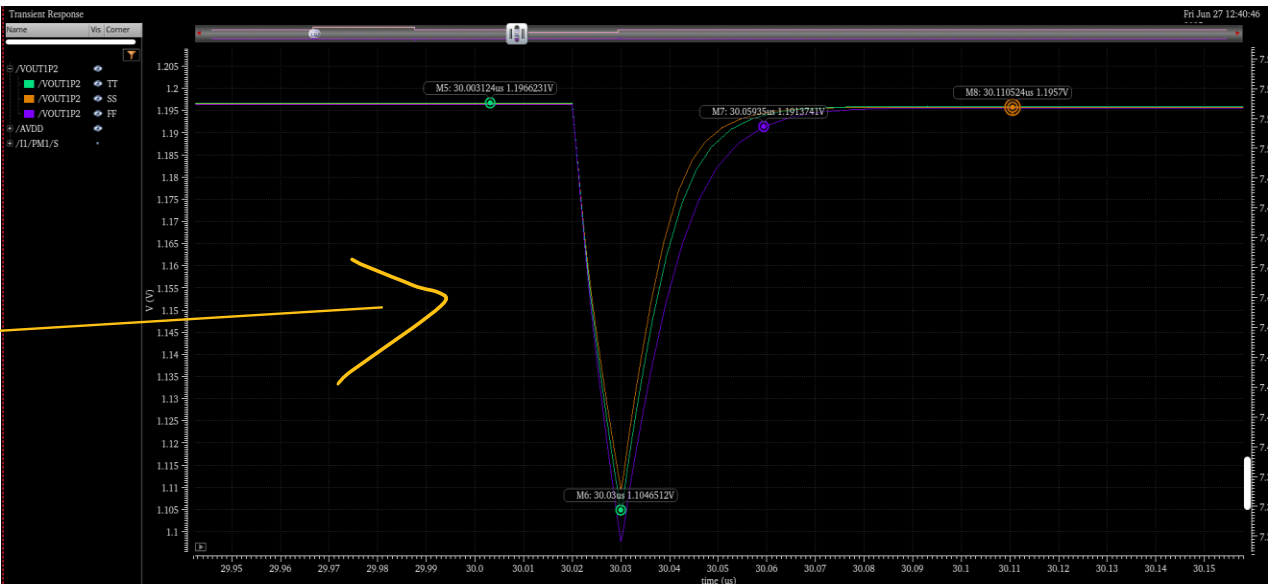
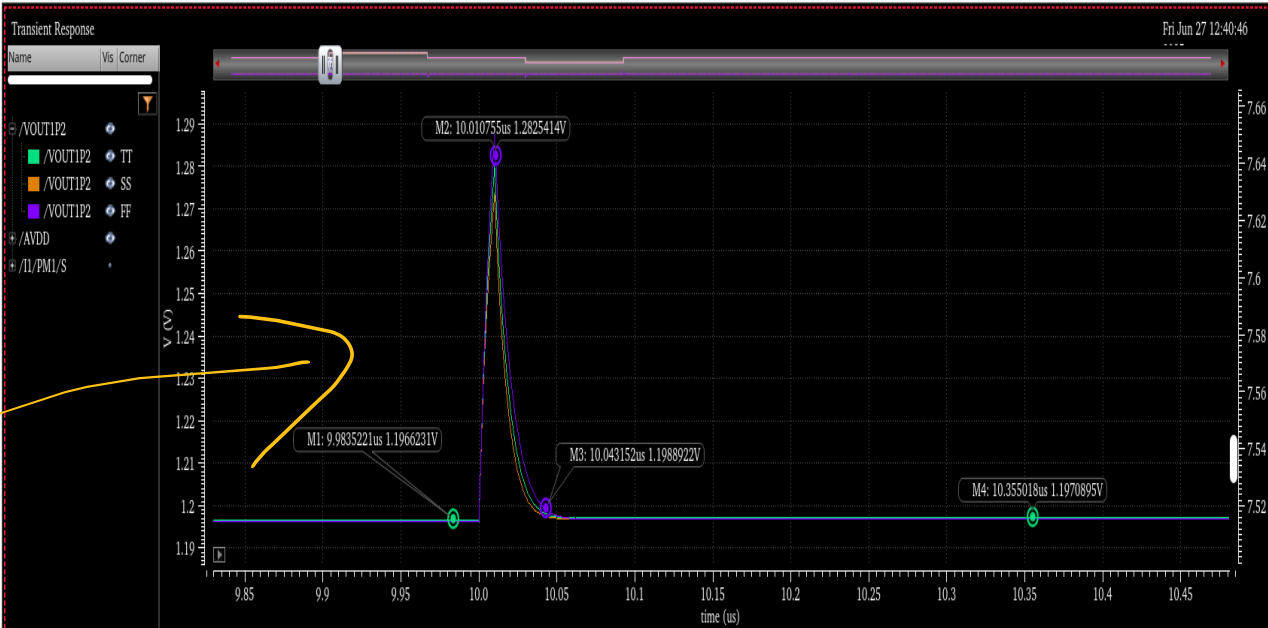
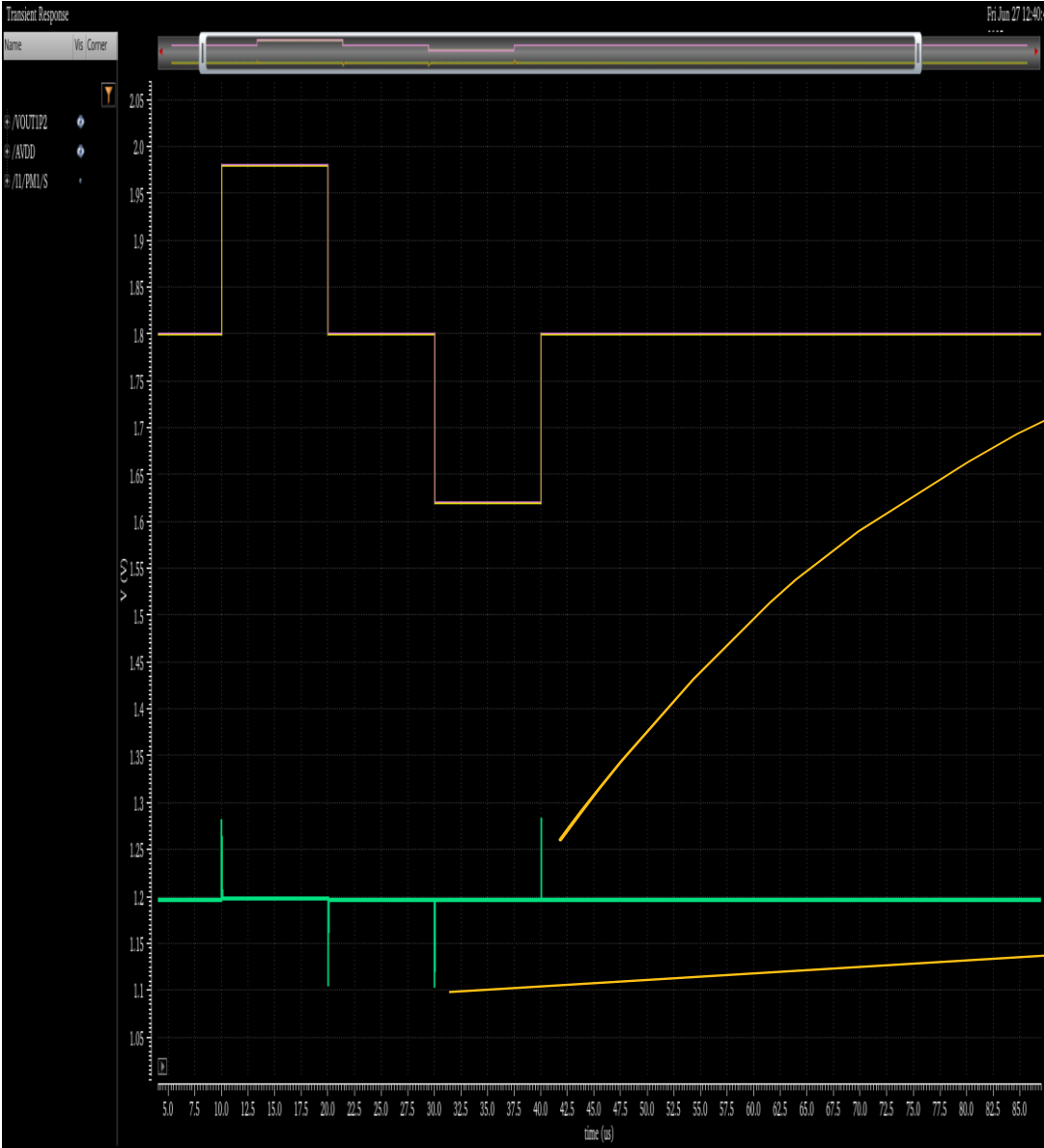
SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
LINE REGULATION(%) MIN	3.2147	125	FF	1.98
LINE REGULATION(%) MAX	3.3727	25	TT	1.8
LINE REGULATION(%) TYPICAL	3.3727	25	TT	1.8

LINE REGULATION : NO LOAD

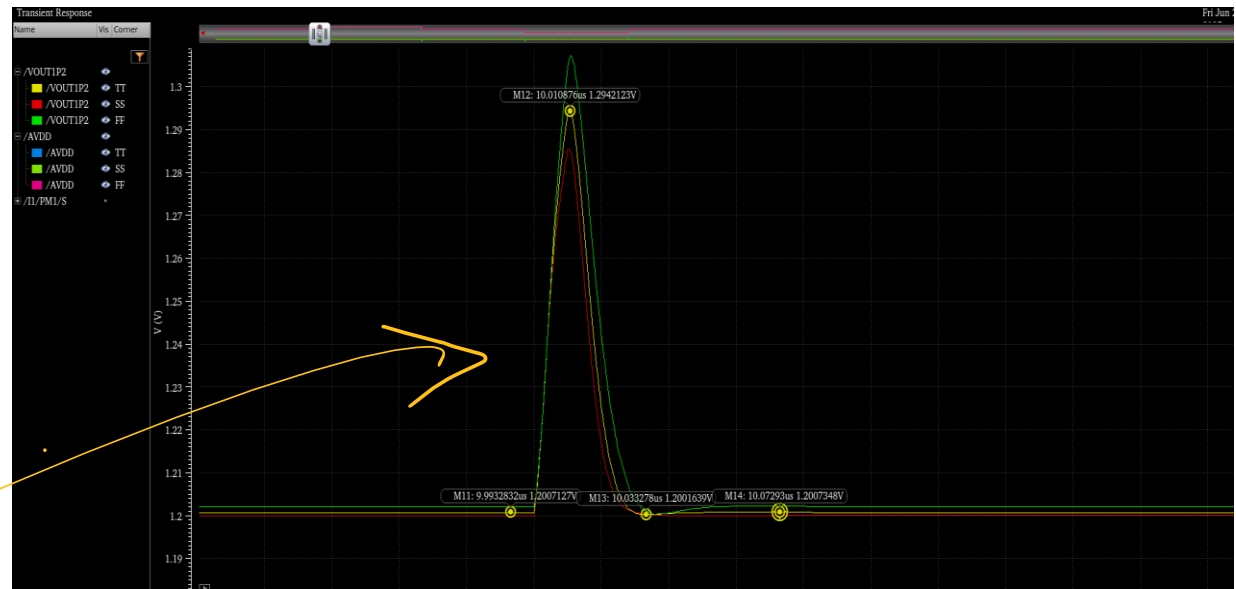
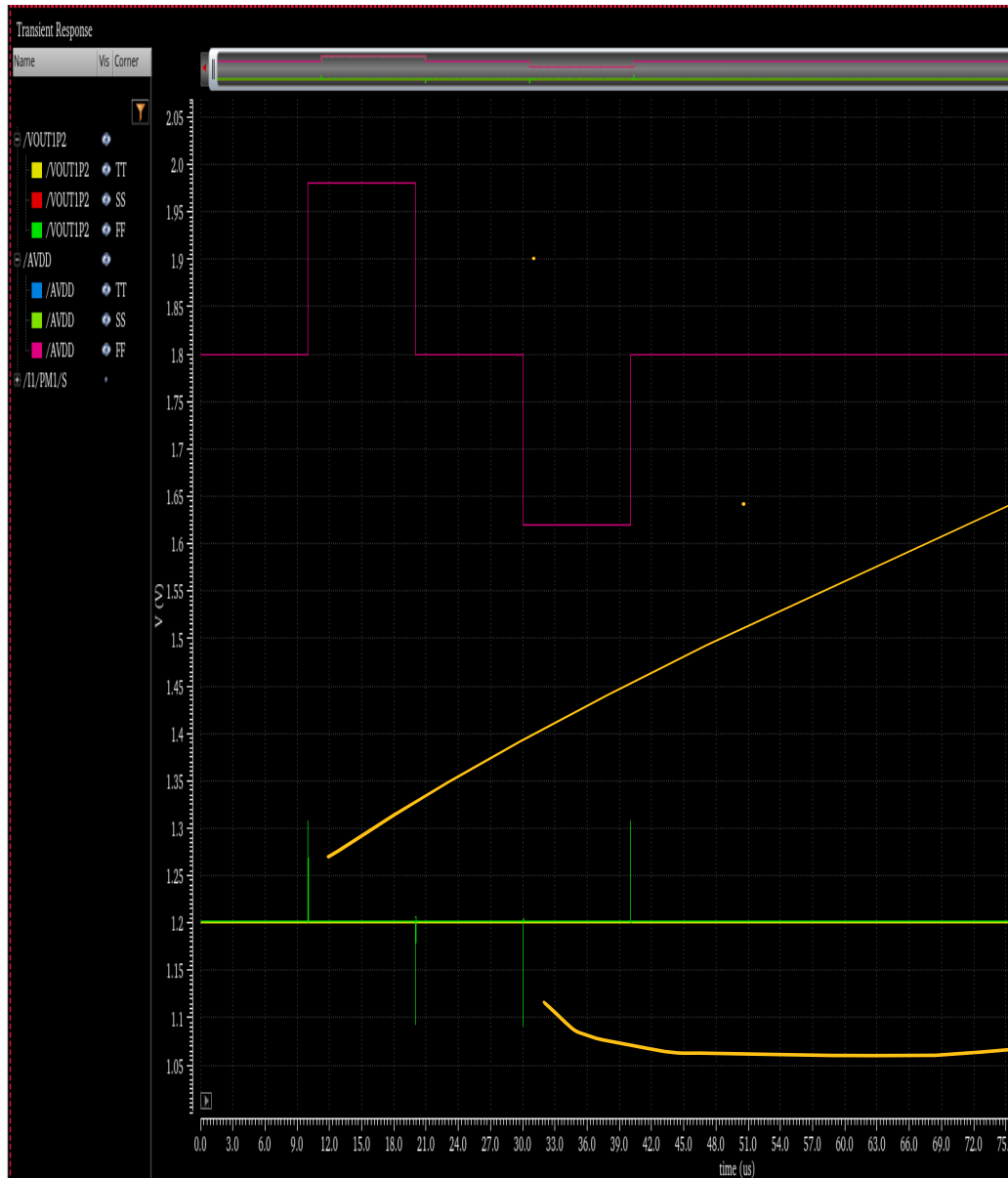


SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
LINE REGULATION(%) MIN	0.003673	25	TT	1.8
LINE REGULATION(%) MAX	0.02039	125	FF	1.98
LINE EGULATION(%) TYPICAL	0.003673	25	TT	1.8

LINE TRANSIENT ANALYSIS FULL LOAD



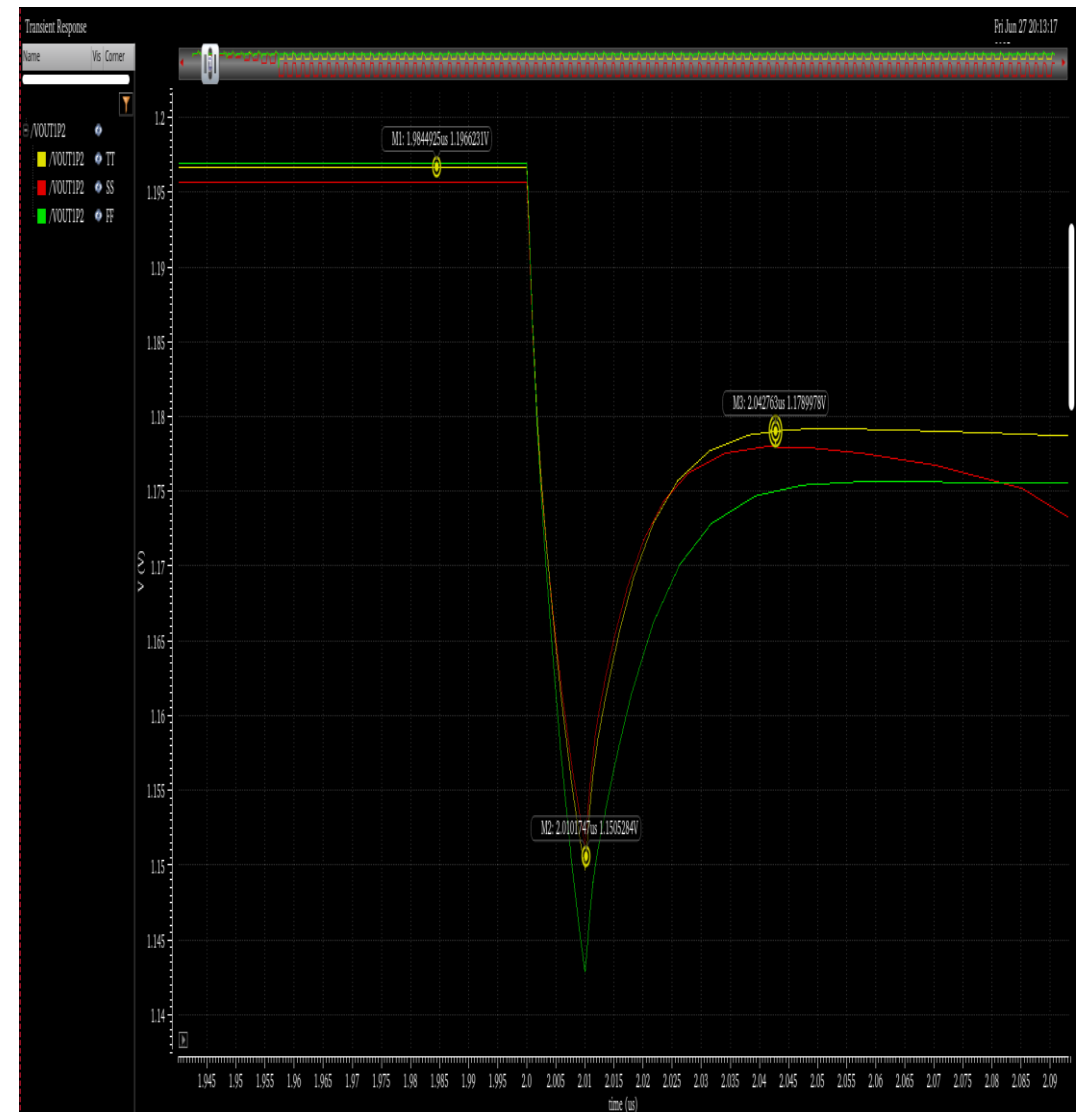
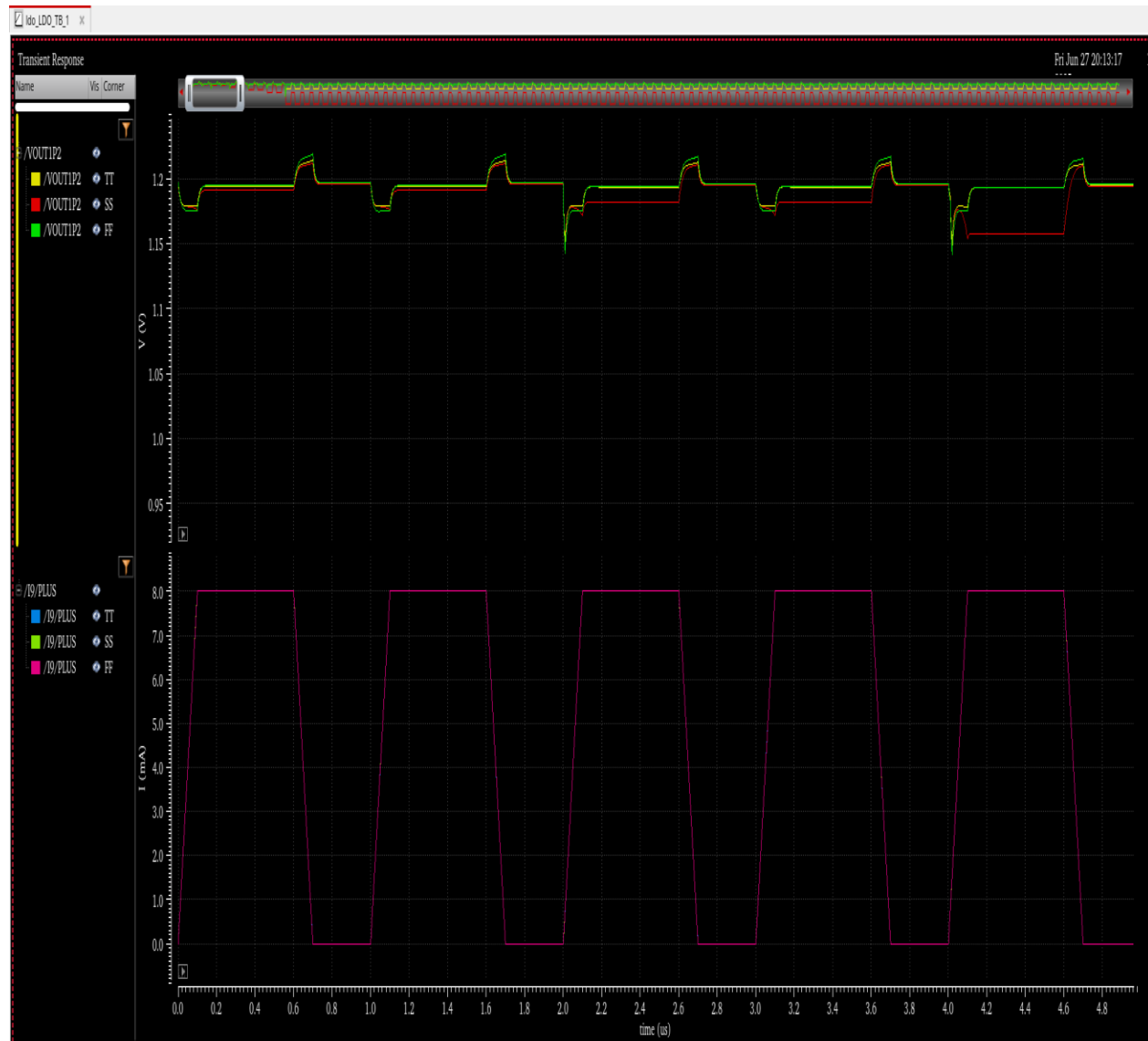
LINE TRANSIENT ANALYSIS NO LOAD



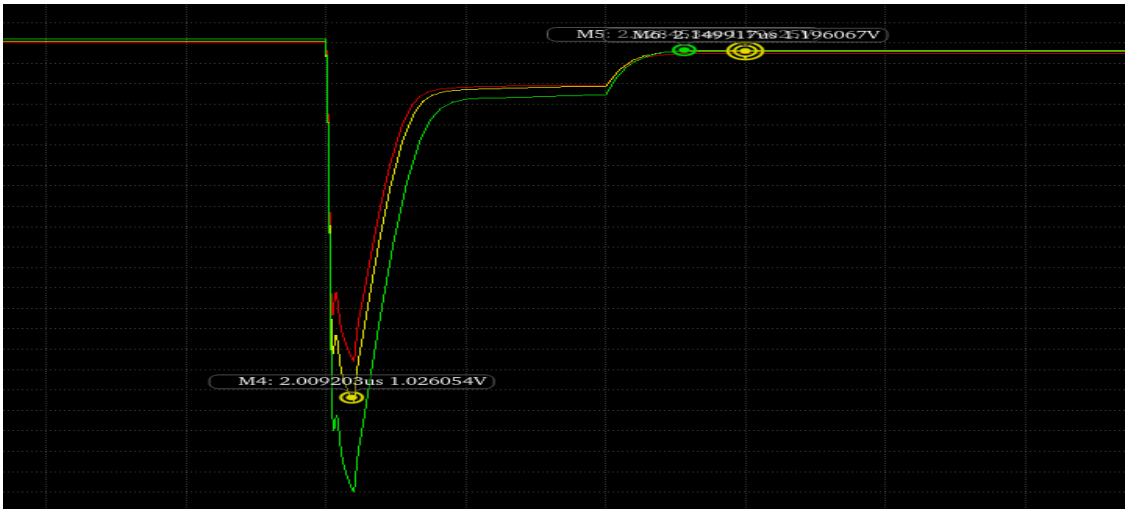
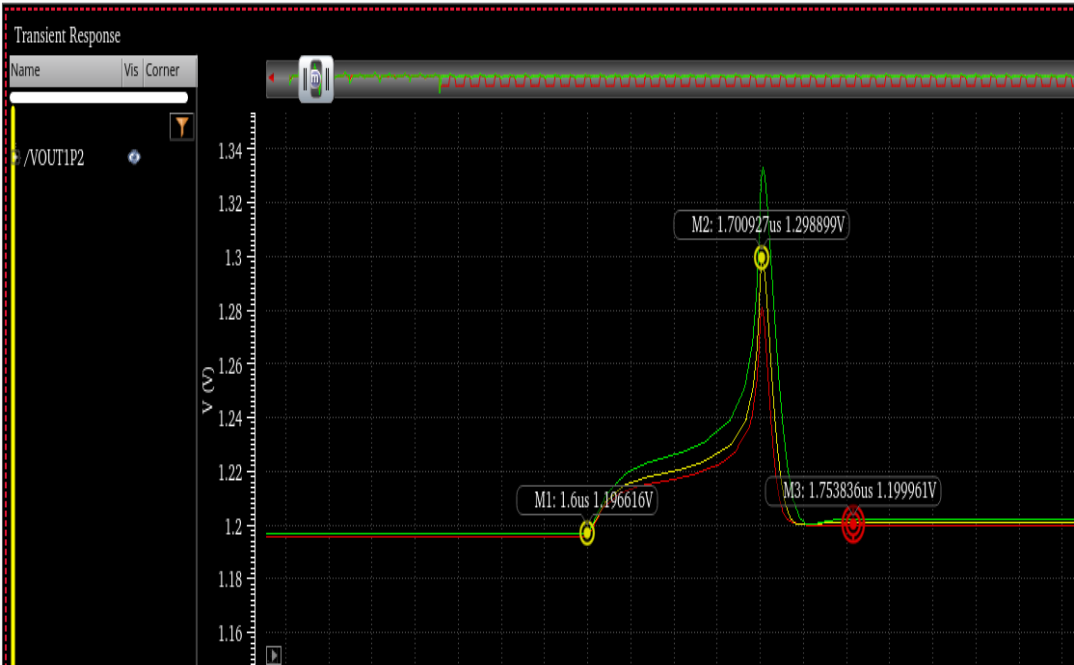
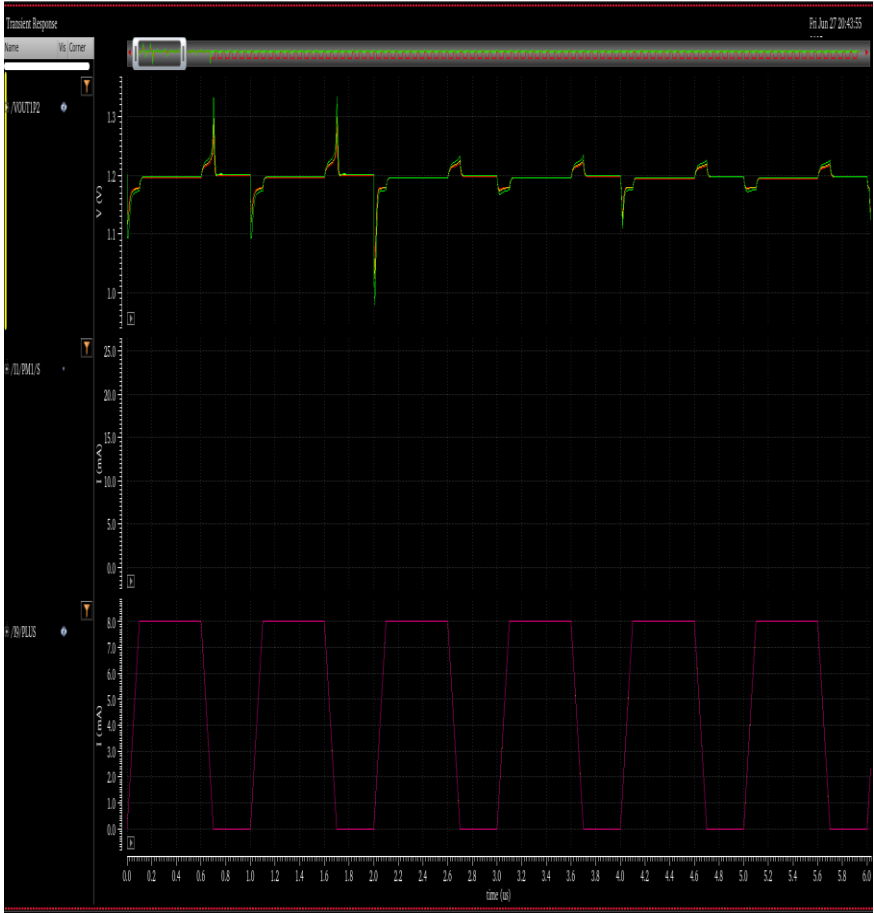
LOAD REGULATION

SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
LOAD REGULATION(%) MIN	0.0106	-40	SS	1.62
LOAD REGULATION(%) MAX	0.0167	125	FF	1.98
LOAD REGULATION(%) TYPICAL	0.012	25	TT	1.8

LOAD REGULATION



LOAD TRANSIENT ANALYSIS



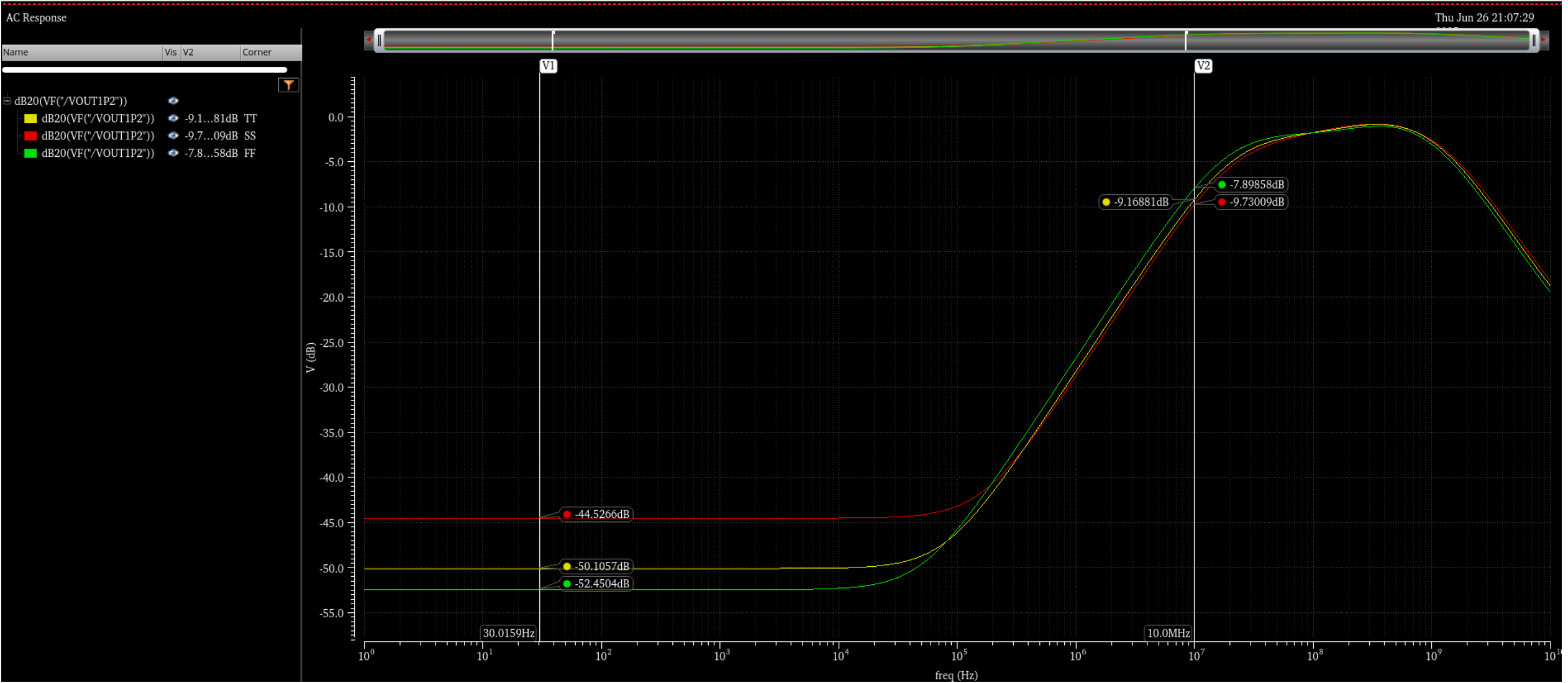
DC PSRR IN FULL LOAD

SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
PSRR(dB)min	-52.450	125	FF	1.98
PSRR(dB)max	-44.526	-40	SS	1.62
PSRR(dB)typical	-50.1057	25	TT	1.8

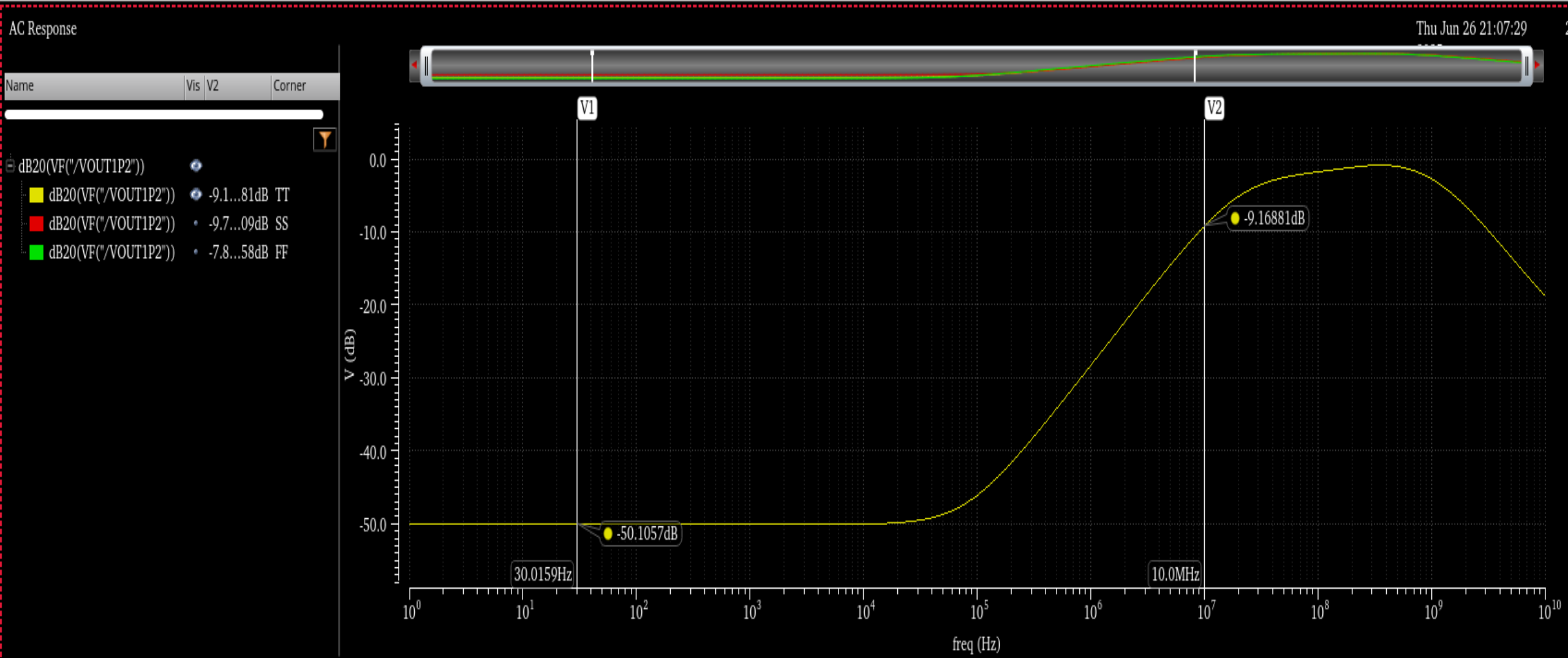
PSRR IN FULL LOAD AT 10MHz

SPECS	VALUES	TEMP	CORNER	SUPPLY VOLTAGE
PSRR(dB)min	-9.73009	-40	SS	1.62
PSRR(dB)max	-7.8985	125	FF	1.98
PSRR(dB)typical	-9.1688	25	TT	1.8

PSRR FULL LOAD



PSRR FULL LOAD(TYPICAL)



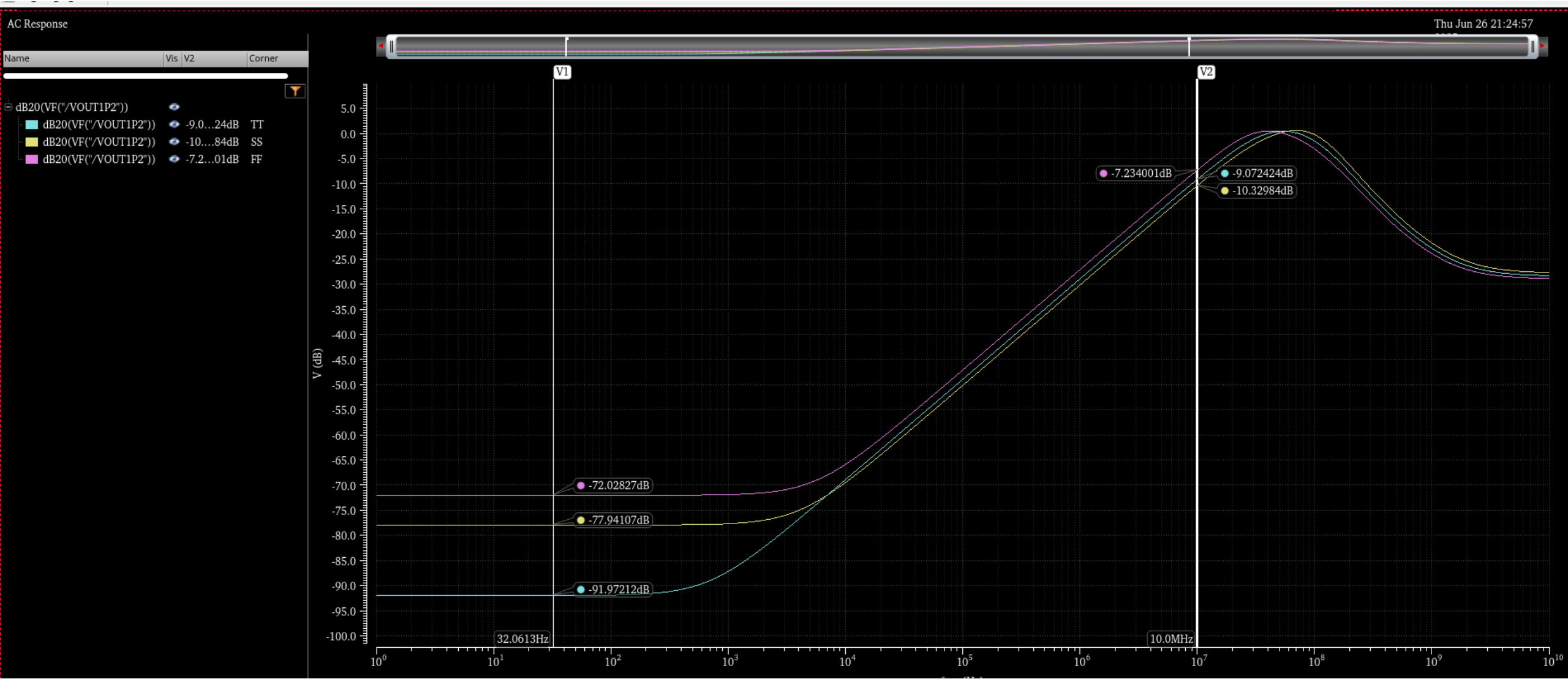
DC PSRR IN NO LOAD

SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
PSRR(dB)min	-91.9721	25	TT	1.8
PSRR(dB)max	-72.0287	125	FF	1.98
PSRR(dB)typical	-91.9721	25	TT	1.8

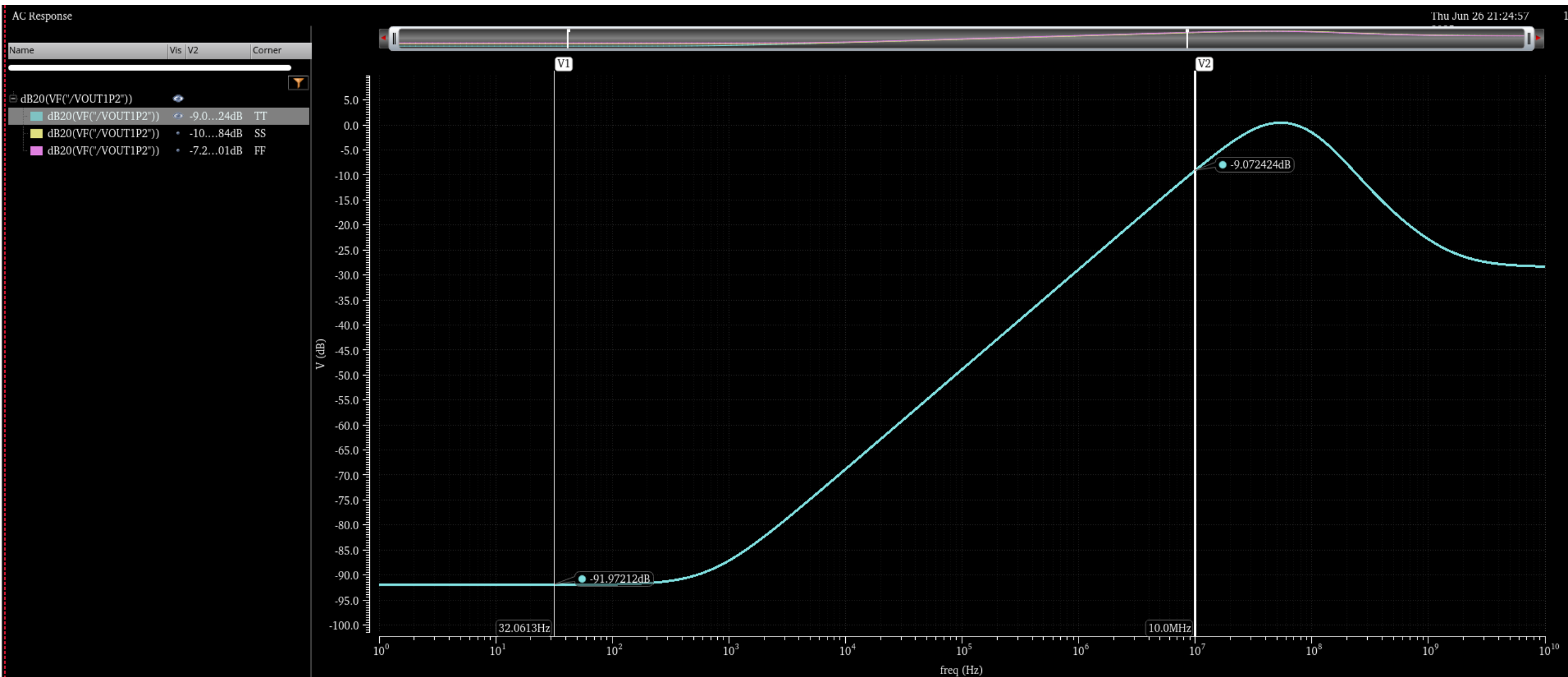
PSRR IN NO LOAD AT 10MHz

SPECS	VALUE	TEMP	CORNER	SUPPLY VOLTAGE
PSRR(dB)min	-10.3298	-40	SS	1.62
PSRR(dB)max	-7.2340	125	FF	1.98
PSRR(dB)typical	-9.0724	25	TT	1.8

PSRR NO LOAD



PSRR NO LOAD(TYPICAL)



TRANSIENT ANALYSIS

